spinning characteristic that turns out to be twice that of the matter particles. They all have “spin-1.”

What about gravity? Well, even before string theory, physicists were able to determine what spin the hypothesized graviton must have to be the transmitter of the gravitational force. The answer: twice the spin of photons, weak gauge bosons, and gluons—i.e., “spin-2.”

In the context of string theory, spin—just like mass and force charges—is associated with the pattern of vibration that a string executes. As with point particles, it’s a bit misleading to think of the spin carried by a string as arising from its spinning literally around in space, but this image does give a loose picture to have in mind. By the way, we can now clarify an important issue we encountered earlier. In 1974, when Scherk and Schwarz proclaimed that string theory should be thought of as a quantum theory incorporating the gravitational force, they did so because they had found that strings necessarily have a vibrational pattern in their repertoire that is massless and has spin-2 —the hallmark features of the graviton. Where there is a graviton there is also gravity.

With this background on the concept of spin, let’s now turn to the role it plays in revealing the loophole in the Coleman-Mandula result concerning the possible symmetries of nature, mentioned in the preceding section.

Supersymmetry and Superpartners

As we have emphasized, the concept of spin, although superficially akin to the image of a spinning top, differs in substantial ways that are rooted in quantum mechanics. Its discovery in 1925 revealed that there is another kind of rotational motion that simply would not exist in a purely classical universe.

This suggests the following question: just as ordinary rotational motion allows for the symmetry principle of rotational invariance (”physics treats all spatial orientations on an equal footing”), could it be that the more subtle rotational motion associated with spin leads to another possible symmetry of the laws of nature? By 1971 or so, physicists showed that the answer to this question was yes. Although the full story is quite involved, the basic idea is that when spin is considered, there is precisely one more symmetry of the laws of nature that is mathematically possible. It is known as supersymmetry.